

JOBSHEET PRAKTIKUM

Mata Kuliah: Pemrograman Web Lanjut

Pertemuan 14 – Implementasi Relation pada Filament (HasMany)

Capaian Pembelajaran

Setelah mengikuti praktikum ini, mahasiswa mampu:

1. Memahami konsep **relationship pada Laravel dan Filament**
2. Menggunakan method **relationship()** pada form Filament
3. Mengimplementasikan **searchable relationship dropdown**
4. Menampilkan data relasi pada tabel Filament
5. Membuat **Relationship Manager**
6. Mengelola relasi **HasMany** pada Filament Admin Panel

Framework yang digunakan:

Filament

Framework backend:

Laravel

A. Latar Belakang

Dalam aplikasi web, sering ditemukan hubungan antar tabel database.

Contoh kasus:

- Satu **Category** memiliki banyak **Post**
- Satu **Post** hanya memiliki satu **Category**

Relasi ini disebut:

- Category → HasMany → Post
Post → BelongsTo → Category

Pada Filament, relationship dapat digunakan untuk:

- Mengisi dropdown otomatis
 - Menampilkan data relasi
 - Mengelola data relasi langsung dari resource
-

B. Relasi Category dan Post

Struktur database sederhana:

categories

id
name
slug

posts

id
title
slug
category_id

Relasi:

Category

↓

HasMany

↓

Post

📌 C. Implementasi Relationship pada Form

Tambahkan kode dan comment kode preload

```
PraktikumPWL > app > Filament > Admin > Resources > Posts > Schemas > PostForm.php > PHP Intelephense > PostForm > configure
21 class PostForm
22 {
23     public static function configure(Schema $schema): Schema
24     {
25         TextInput::make("title")
26             ->rules('required | min:3 | max:10'),
27         TextInput::make("slug")
28             ->rules('required')
29             ->unique()
30             ->validationMessages([
31                 "unique" => "Slug must be unique"
32             ]),
33         Select::make("category_id")
34             ->relationship("category", "name")
35             ->options(Category::all()->pluck("name", "id"))
36             ->required()
37             // ->preload()
38             ->searchable(),
39         ColorPicker::make("color"),
40     }
41 }
```

Parameter	Fungsi
category	nama relasi pada model
name	field yang ditampilkan

📌 D. Membuat Dropdown Searchable

Jika kategori sangat banyak, gunakan fitur pencarian:

- `Select::make('category_id')`
 - >`relationship('category','name')`
 - >`searchable()`

```
PraktikumPWL > app > Filament > Admin > Resources > Posts > Schemas > PostForm.php > PHP Intelephense > PostForm > configure
21  class PostForm
23      public static function configure(Schema $schema): Schema
35          TextInput::make("title")
36              ->rules('required | min:3 | max:10'),
37          TextInput::make("slug")
38              ->rules('required')
39              ->unique()
40              ->validationMessages([
41                  "unique" => "Slug must be unique"
42              ]),
43          Select::make("category_id")
44              ->relationship("category", "name")
45              ->options(Category::all()->pluck("name", "id"))
46              ->required()
47              // ->preload()
48              ->searchable(),
49          ColorPicker::make("color"),
```

◆ Hasil

Dropdown kategori:

- Bisa dicari
- Lebih cepat untuk dataset besar
- Tidak perlu memuat semua data sekaligus

Posts > Create

Create Post

Post Details
Fill in the details of the post

Title

Slug

Category*
Select an option
php
PHP

Color

Image Upload
Image*
Drag & Drop your files or **Browse**

Meta Information
Tags
New tag
Published ☐
Published at
dd/mm/yyyy --:--:--

Create Create & create another Cancel

✦ E. Relationship pada Model

Pastikan relasi sudah dibuat pada model.

Model Post

- public function category()
{
 return \$this->belongsTo(Category::class);
}

PraktikumPWL > app > Models > Post.php > PHP Intelephense > Post

```
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Model;
6
7  class Post extends Model
8  {
9      //
10     protected $fillable = [
11         'title',
12         'slug',
13         'category_id',
14         'color',
15         'image',
16         'body',
17         'tags',
18         'published',
19         'published_at',
20     ];
21     protected $casts = [
22         'tags' => 'array',
23         'published' => 'boolean',
24         'published_at' => 'date',
25     ];
26
27     public function category()
28     {
29         return $this->belongsTo(Category::class);
30     }
31
32 }
33
34
```

Model Category

```
public function posts()
{
    return $this->hasMany(Post::class);
}
```

PraktikumPWL > app > Models > Category.php > PHP Intelephense > Category

```
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Model;
6
7  class Category extends Model
8  {
9      //
10     protected $fillable = [
11         "name",
12         "slug"
13     ];
14     public function posts()
15     {
16         return $this->hasMany(Post::class);
17     }
18
19 }
20
```

✦ F. Menampilkan Data Relasi pada Table

Pada Post Table:

- `TextColumn::make('category.name')`

PraktikumPWL > app > Filament > Admin > Resources > Posts > Tables > PostsTable.php > PHP Intelephense > PostsTable > configure

```
22  class PostsTable
23  {
24      public static function configure(Table $table): Table
25      {
26          $table->sortable()
27              ->toggleable()
28              ->searchable(),
29          TextColumn::make('slug')
30              ->sortable()
31              ->toggleable()
32              ->searchable(),
33          TextColumn::make('category.name')
34              ->sortable()
35              ->toggleable()
36              ->searchable(),
37          ColorColumn::make('color'), ...
38          ->disk('public'), ...
39      }
40  }
41
42
43
44
45
```

◆ Hasil

Tabel akan menampilkan:

Title	Category
Post Laravel	Laravel
Post PHP	PHP

📌 G. Membuat Relationship Manager

- Filament menyediakan fitur **Relationship Manager** untuk mengelola relasi langsung dari resource.
-

◆ Jalankan Command

- `php artisan make:filament-relation-manager`

Isi parameter:

- Resource : `CategoryResource`
Relationship : `posts`
Title Column : `title`

```
PS C:\laragon\www\PraktikumPwL> php artisan make:filament-relation-manager CategoryResource posts title

Filament can link this to an existing resource, which will open the resource's pages instead of models when links are clicked. It will also inherit the resource's configuration.

Do you want to do this? (yes/no) [no]
> no

Would you like to use an existing form schema class? (yes/no) [no]
> no

Should the configuration be generated from the current database columns? (yes/no) [no]
> no

Should there be a read-only "view" modal on the relation manager? (yes/no) [no]
> no

Would you like to use an existing table class? (yes/no) [no]
> no

Does the model use soft-deletes? (yes/no) [no]
> no

What type of relationship is this?
HasMany ..... Illuminate\Database\Eloquent\Relations\HasMany
BelongsToMany ..... Illuminate\Database\Eloquent\Relations\BelongsToMany
MorphToMany ..... Illuminate\Database\Eloquent\Relations\MorphToMany
MorphToMany ..... Illuminate\Database\Eloquent\Relations\MorphToMany
Other ..... other
> HasMany
HasMany

[INFO] Filament relation manager [App\Filament\Admin\Resources\Categories\RelationManagers\PostsRelationManager] created successfully.

[INFO] Make sure to register the relation in [App\Filament\Admin\Resources\Categories\CategoryResource::getRelations()].

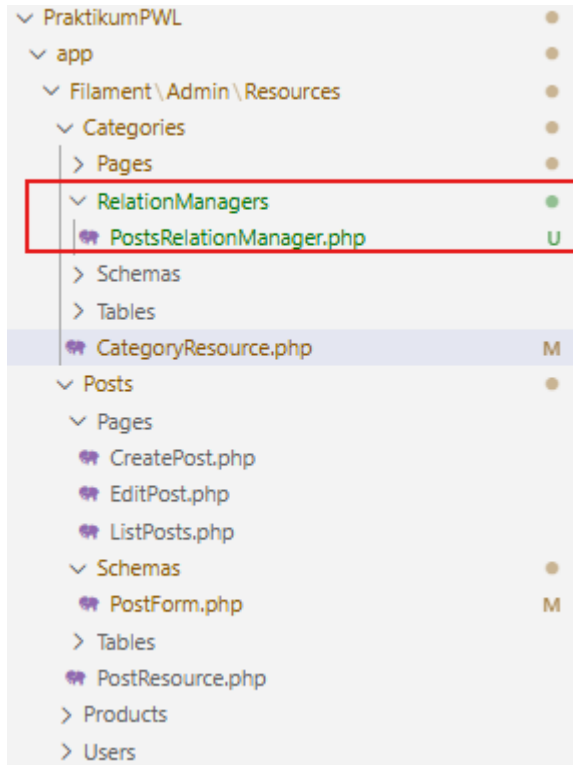
PS C:\laragon\www\PraktikumPwL>
```

◆ File yang Dibuat

CategoryResource

└─ RelationManagers

└─ PostsRelationManager.php



✦ H. Menghubungkan Relationship Manager

Buka:

- CategoryResource.php

Tambahkan pada method:

- ```
public static function getRelations(): array
{
 return [
 PostsRelationManager::class,
];
}
```

PraktikumPWL > app > Filament > Admin > Resources > Categories > CategoryResource.php

```
5 use App\Filament\Admin\Resources\Categories\Pages\CreateCategory;
6 use App\Filament\Admin\Resources\Categories\Pages>EditCategory;
7 use App\Filament\Admin\Resources\Categories\Pages>ListCategories;
8 use App\Filament\Admin\Resources\Categories\Schemas\CategoryForm;
9 use App\Filament\Admin\Resources\Categories\Tables\CategoriesTable;
10 use App\Models\Category;
11 use Filament\Resources\Resource;
12 use Filament\Schemas\Schema;
13 use Filament\Tables\Table;
14 use App\Filament\Admin\Resources\Categories\RelationManagers\PostsRelationManager;
15
16 class CategoryResource extends Resource
17 {
18 protected static ?string $model = Category::class;
19
20 // protected static string|BackedEnum|null $navigationIcon = Heroicon::OutlinedRectangleStack;
21
22 protected static ?string $recordTitleAttribute = 'name';
23
24 public static function getNavigationIcon(): string|null
25 { ... }
26
27 public static function form(Schema $schema): Schema
28 { ... }
29
30 public static function table(Table $table): Table
31 { ... }
32
33 public static function getRelations(): array
34 {
35 return [
36 PostsRelationManager::class
37];
38 }
39
40 public static function getPages(): array
41 { ... }
42 }
43
44
45
46
47
48
49
50
51
52
53
54
```

Buka model -> Category.php dan tambahkan kode berikut



```

PraktikumPWL > app > Models > Category.php > ...
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Category extends Model
8 {
9 //
10 protected $fillable = [
11 "name",
12 "slug"
13];
14 public function posts()
15 {
16 return $this->hasMany(Post::class);
17 }
18
19 }

```

## 📌 I. Hasil Relationship Manager

Saat membuka **Edit Category**, akan muncul tabel Post yang berelasi.

Fitur yang tersedia:

- View Post
- Create Post
- Edit Post
- Delete Post

Categories > laravel > Edit

Delete

Name\*

laravel

Slug\*

laravel

Save changes

Cancel

Posts

New postAssociate

Q Search

☐ Title

☐ 1122

Edit × Dissociate Delete

☐ Filament

Edit × Dissociate Delete

Showing 1 to 2 of 2 results

Per page 10

---

## ✦ J. Menambahkan Kolom pada Relationship Table

Contoh:

- `TextColumn::make('title'),`  
`TextColumn::make('slug'),`  
`TextColumn::make('created_at'),`

```
PraktikumPWL > app > Filament > Admin > Resources > Categories > RelationManagers > PostsRelationManager.php
20 {
 windsurf: Refactor | Explain | Generate Function Comment | X
23 public function form(Schema $schema): Schema
24 {
25 return $schema
26 ->components([
27 TextInput::make('title')
28 ->required()
29 ->maxLength(255),
30 TextInput::make('slug')
31 ->required()
32 ->maxLength(255),
33]);
34 }
35
36 public function table(Table $table): Table
37 {
38 return $table
39 ->recordTitleAttribute('title')
40 ->columns([
41 TextColumn::make('title')
42 ->searchable(),
43 TextColumn::make('slug')
44 ->searchable(),
45 TextColumn::make('created_at')
46 ->dateTime(),
47])
 }
```

---

### ◆ Hasil

Tabel Post di Category akan menampilkan:

Categories > laravel > Edit

## Edit laravel

**Name\*** laravel **Slug\*** laravel

[Save changes](#) [Cancel](#) [Delete](#)

**Posts** [New post](#) [Associate](#)

Search

| <input type="checkbox"/> Title    | Slug         | Created at           |                                                                        |
|-----------------------------------|--------------|----------------------|------------------------------------------------------------------------|
| <input type="checkbox"/> 1122     | test1        | Mar 1, 2026 10:11:46 | <a href="#">Edit</a> <a href="#">Dissociate</a> <a href="#">Delete</a> |
| <input type="checkbox"/> Filament | filament2026 | Mar 3, 2026 14:32:12 | <a href="#">Edit</a> <a href="#">Dissociate</a> <a href="#">Delete</a> |
| <input type="checkbox"/> figma    | desain       | Mar 5, 2026 08:15:37 | <a href="#">Edit</a> <a href="#">Dissociate</a> <a href="#">Delete</a> |

Showing 1 to 3 of 3 results

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## K. Membuat Form Create Post pada Relationship

Pada Relationship Manager:

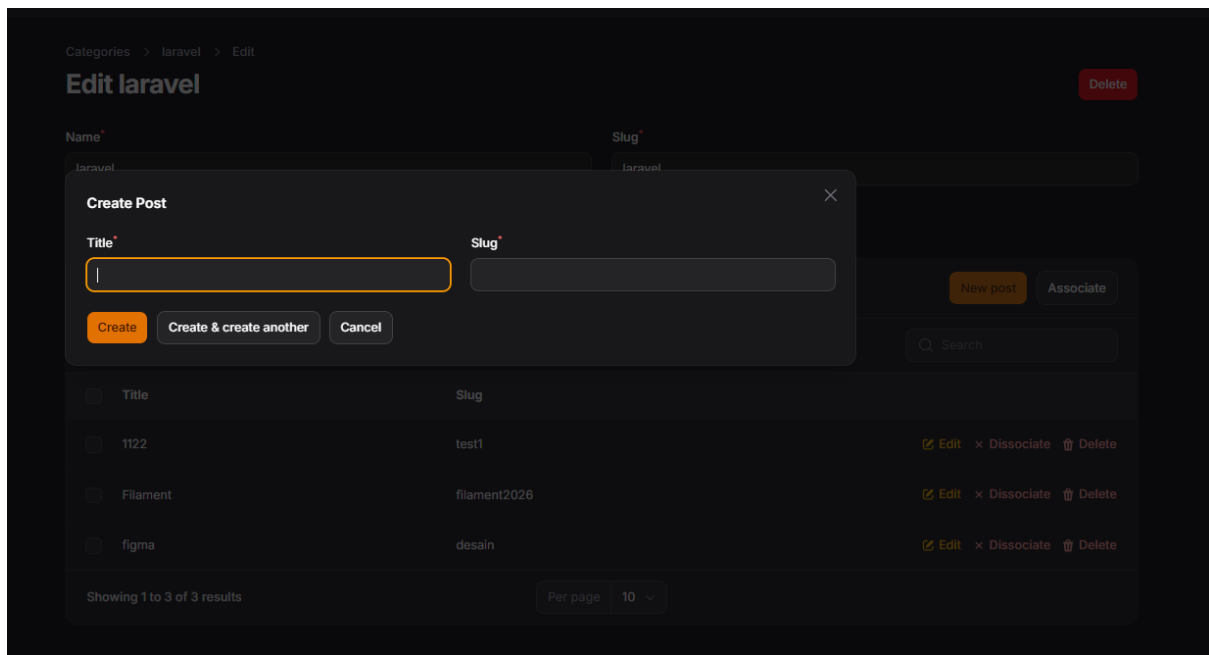
- `TextInput::make('title'),`  
`TextInput::make('slug'),`

```
PraktikumPWL > app > Filament > Admin > Resources > Categories > RelationManagers > PostsRelationManager.php
10 use Filament\Actions\EditAction;
11 use Filament\Forms\Components\TextInput;
12 use Filament\Resources\RelationManagers\RelationManager;
13 use Filament\Schemas\Schema;
14 use Filament\Tables\Columns\TextColumn;
15 use Filament\Tables\Table;
16
17
18
19 class PostsRelationManager extends RelationManager
20 {
21 protected static string $relationship = 'posts';
22
23 public function form(Schema $schema): Schema
24 {
25 return $schema
26 ->components([
27 TextInput::make('title')
28 ->required()
29 ->maxLength(255),
30 TextInput::make('slug')
31 ->required()
32 ->maxLength(255),
33]);
34 }
35 }
```

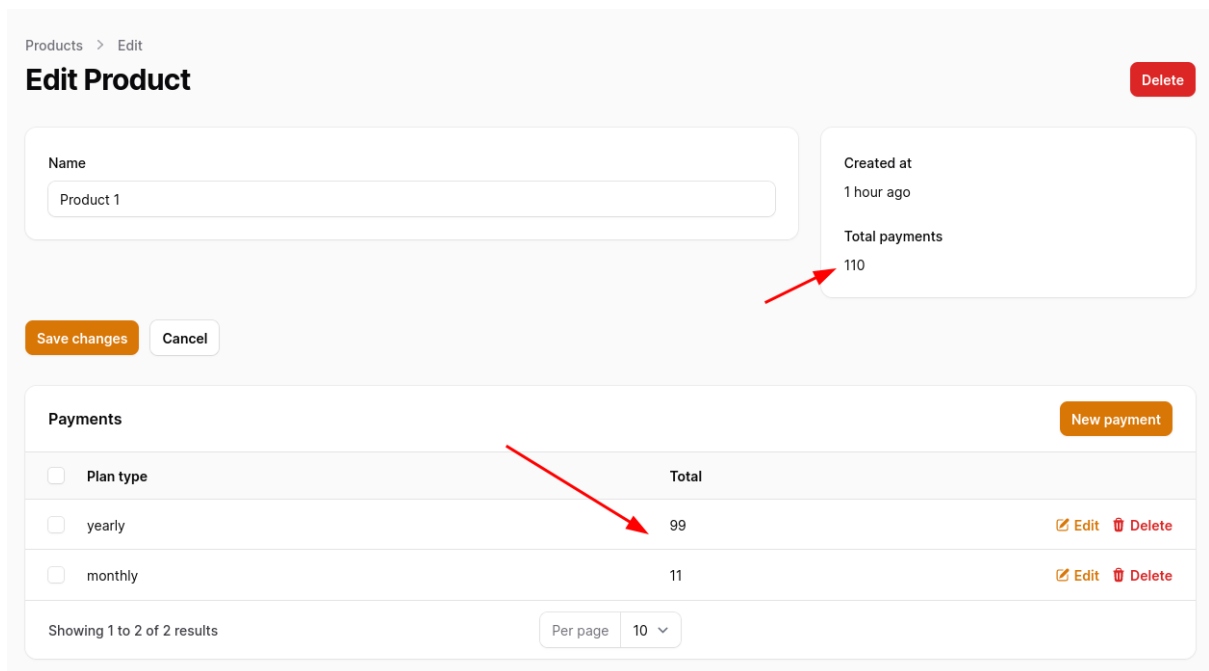
Saat membuat post baru dari kategori:

- `category_id` otomatis terisi

- tidak perlu memilih kategori lagi



## 📌 L. Contoh Tampilan Relationship Manager



## 📌 M. Keuntungan Menggunakan Relationship Manager

| Fitur                        | Manfaat                          |
|------------------------------|----------------------------------|
| Manajemen relasi             | Data relasi lebih mudah dikelola |
| CRUD langsung                | Tidak perlu berpindah halaman    |
| Otomatis isi foreign key     | Mengurangi kesalahan input       |
| Terintegrasi dengan resource | UI lebih konsisten               |

## N. Latihan Praktikum

1. Buat relasi **Category** → **Posts**
2. Implementasikan relationship dropdown pada Post Form
3. Tambahkan fitur **searchable()**
4. Tampilkan kategori pada Post Table
5. Buat **Relationship Manager** pada Category
6. Tambahkan kolom tambahan pada tabel relasi

Screenshot yang harus dikumpulkan:

- Dropdown kategori pada Post Form
- Tabel Post dengan kategori
- Relationship Manager pada Category
- Create Post dari Category

## O. Analisis & Diskusi

1. Apa perbedaan **relationship()** dengan **options()**?
2. Mengapa searchable penting untuk dataset besar?
3. Apa fungsi Relationship Manager pada Filament?
4. Kapan menggunakan **HasMany** dan **BelongsTo**?

## P. Kesimpulan

Pada praktikum ini mahasiswa telah mempelajari:

- Konsep relationship pada Laravel
- Implementasi relationship pada Filament form
- Penggunaan dropdown relationship

- Penggunaan Relationship Manager
- Pengelolaan data relasi pada admin panel